Two homology covers in rank four

Author manuscript, phase256, 11 October 2026. This candidate proof is prepared for independent review and is not yet admitted. Its conclusion concerns finite groups of rank four. It does not answer the finite-cover question in higher rank. The earlier arbitrary-cover theorem in phase216 is separate.

1. Statement and dependencies

A marked string group of rank four is a group generated by distinct nonidentity involutions s_0,s_1,s_2,s_3, with s_i and s_j commuting whenever |i−j|>1. Write G_I=⟨s_i:i∈I⟩. Throughout this manuscript G is finite and

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s_0 ∉ G123,                  s_3 ∉ G012.                  (1)
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No intersection property is assumed. Put

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g=|G|, A=G012, B=G123, C=G12, q=ord(s_1s_2).
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The distinctness assumptions imply q≥2 and |G02|=|G03|=|G13|=4.

Theorem 1 (candidate). For every prime p>4g there is a finite marked string C-group T mapping onto G whose kernel is a p-group.

A finite polytope's monodromy group satisfies (1): the subgroup generated by all marks except s_i preserves the rank-i face of every flag, whereas s_i changes that face. Thus Theorem 1 would imply the following conclusion.

Corollary 2 (candidate). Every finite rank-four polytope whose monodromy group is not a string C-group has infinitely many pairwise nonisomorphic finite minimal regular polytopal covers. For every prime p>4|Mon(P)|, one can choose such a cover whose degree over Mon(P) is a nontrivial power of p.

Problem 5 in Cunningham–Pellicer, *Open problems on k-orbit polytopes*, §3.1, asks whether non-C monodromy can coexist with a unique minimal regular polytopal cover. The question has no finiteness restriction. Cunningham, *Chiral polytopes whose smallest regular cover is a polytope*, §4, expresses the same expectation and a more specific family conjecture. Corollary 2 asserts a finite-cover strengthening for finite polytopes of rank four; it is not a claim to settle the finite strengthening in all ranks.

The construction has two steps. First take the mod-p homology cover of a surface with boundary; this repairs the two rank-three ends. Then fill the middle boundary polygons and take another mod-p homology cover. The new argument proves the precise complement conditions needed in this second step, including the nonorientable case.

The proof reuses the cellular intersection argument and closed-surface homology-cover theorem of phase211, §§3–5. Those arguments are restated at the point where they are used. The raw surface and its relative collapses were developed in the unfinished phase224_finite_rank_four.md, §§2–5; this manuscript supplies the previously missing uniform second step. The known existence of finite regular polytopal covers is due to Monson–Schulte. Mere existence is not the new conclusion asserted here. The new assertion is the existence of a cover with a p-group kernel for every sufficiently large prime, and its consequence for finite minimal covers.

2. The surface with boundary and the first cover

Let W be generated by r_0,r_1,r_2,r_3, with defining relations r_i²=1 and [r_0,r_2]=[r_0,r_3]=[r_1,r_3]=1. Let K=ker(W→G).

Construct a polygonal complex X with vertex set G and a colored edge from x to xs_i for every i. Attach one square for each left coset of G02, G03, and G13 along its alternating two-color cycle. The squares

have embedded boundaries. Each edge of color 0 or 3 lies in two squares; each edge of color 1 or 2 lies in one. The link of every vertex is the path

2 − 0 − 3 − 1.

Consequently X is a connected compact surface with boundary. Its boundary circles are exactly the alternating 12 cycles, one for each left coset of C. Its cell counts give

χ(X)=g−2g+3g/4=−g/4.

The path presentation gives π_1(X,1)=K: backtracking gives the involution relations, and square homotopies give all conjugates of the three commuting relations. One square per coset suffices, since its boundary based at another corner is a conjugate of that boundary or its inverse.

For I⊆{0,1,2,3}, let X_I be the connected colored subcomplex through 1, including exactly the available squares whose colors belong to I. Its vertex set is G_I. Set X_A=X012 and X_B=X123.

Fix a prime p and put

V=H_1(X;F_p), Φ=[K,K]K^p, H=W/Φ.

The subgroup Φ is characteristic in K and hence normal in W. The surface cover X_p→X with deck group V is connected and has the Cayley graph of H as its one-skeleton. Since a compact connected surface with nonempty boundary has H_2=0 over every field,

dim V=1+g/4, |H|=g p^(1+g/4). (2)

All marks and (1) lift to H. Write A_H=H012, B_H=H123, and C_H=H12.

Lemma 3. The groups A_H and B_H, with their indicated marks, are string C-groups.

Proof. For S=X01∪X12, every 02 square has a 2-edge outside S: otherwise s_0∈G12. Collapse that square across such an edge; each 2-edge belongs to only one square. Every 13 square has a 1-edge outside S: all vertices of S belong to A, so two such edges in S would imply s_3∈A. Collapse these squares too. Each remaining 03 square has a 0-edge outside S, for the same reason. The 02 square formerly incident with that edge was already removed, so it is now a free edge and the 03 square can be collapsed across it. These collapses fix S and leave only a graph attached to S. Therefore H_1(S;F_p)→H_1(X;F_p) is injective. The dual argument applies to X12∪X23.

Here is the homological intersection argument used for these pairs. For connected subcomplexes E,F with a common base point, suppose H_1(E∩F)=0 and H_1(E∪F)→H_1(X) is injective. If paths in E and F lift from the chosen base point to the same vertex, their difference is zero in H_1(X), hence in H_1(E∪F). The Mayer–Vietoris connecting map first shows that their common projected endpoint belongs to the base component C_0 of E∩F. Comparing both paths with a path in C_0 and using H_1(E∩F)=0 then shows that they have the same lifted endpoint as that path. Thus the chosen lifted components intersect exactly in the chosen lift of C_0.

For the two pairs above, E∩F is a disjoint union of single edges, so this argument gives H01∩H12=H1 and H12∩H23=H2. A group generated by two distinct involutions has the rank-two intersection property. The recursive criterion for string C-groups now proves the lemma. ■

3. Removing an end and detecting boundary classes

We first record two elementary connectivity facts with their cell details.

Lemma 4. The induced Cayley graph on G\A is connected. The same holds for G\B, and for the corresponding ends in any finite marked group satisfying (1).

Proof. Contract each left A-coset, retaining the edges of color 3 between cosets. The resulting finite graph is connected and vertex-transitive under left multiplication by G. It has at least two vertices. A

finite connected vertex-transitive graph has no cut vertex: a finite connected graph with at least two vertices has a noncut vertex (for example, a leaf of a spanning tree), and transitivity carries this property to every vertex. Deleting the vertex A therefore leaves a connected graph. Every remaining A-coset is internally connected using colors 0,1,2, so a path in the contracted graph lifts to a path avoiding A. The dual argument proves the other assertion. Loops or repeated edges in the contracted graph do not affect this argument. ■

Lemma 5. The following complements have the same components as their induced Cayley graphs on vertices outside the indicated subcomplexes:

(a) X\X_A and X\X_B;

(b) Y(Y012∪Y123), where Y is obtained from X by filling every middle boundary circle with a disk.

Here the analogous statement holds for any finite marked group satisfying (1).

Proof. An outside face always has an outside vertex, and the outside vertices of that face are connected along its boundary through outside vertices. We check this assertion explicitly. For X_A, a 02 square is wholly in X_A or has no vertex in A. A 03 square meets A in one 0-edge or not at all; a 13 square meets it in one 1-edge or not at all. These are all possibilities because s_3∉A. Thus the asserted boundary connectivity holds. The argument for X_B is dual.

For the union of the two ends, a 02 square lying outside Y012 has no A vertex; it meets B in one 2-edge or not at all because s_0∉B. A 13 square is dual. A 03 square meets A in at most one 0-row and B in at most one 3-column; removing such a row and column leaves either the whole boundary, a connected edge, or a single vertex. A middle 12 disk is wholly in the union or has no vertex there, since C belongs to both ends.

We also need that every outside edge has an outside endpoint. For the single end A, edges of colors 0,1,2 with an A endpoint are selected, whereas an edge of color 3 cannot have both endpoints in A. For the top union, an edge of color 1 or 2 with a selected endpoint is selected. An outside edge of color 0 has no endpoint in A and cannot have both endpoints in B, because s_0∉B. Color 3 is dual. Thus the required assertion holds.

At a vertex outside the selected subcomplex every incident edge and face is outside it. The link of that vertex is connected, including the boundary case in X. Hence all incident outside faces belong to the same component. Conversely, whenever outside faces meet across an outside edge, or around an outside vertex, their outside vertices are connected by the boundary paths just described. Thus the components of the outside face adjacency graph, the topological complement, and the induced outside-vertex graph are in bijection. ■

For a boundary circle xC, choose either orientation and denote its class in V by u_x. To view this as the class of a based loop one may use any path from 1 to x; the path and its inverse cancel in first homology. The choice of orientation only changes the sign. Define

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V_A=image(H_1(X_A;F_p)→V),
V_B=image(H_1(X_B;F_p)→V).
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Lemma 6. For every x∈G,

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u_x∈V_A ⇔ x∈A,              u_x∈V_B ⇔ x∈B.          (3)
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Proof. If x∈A, the boundary circle belongs to X_A and is joined to 1 inside X_A, so u_x∈V_A. Suppose x∉A. By Lemmas 4–5 the complement of X_A is connected, including its outside face adjacency graph. The number of middle boundary circles outside A is

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[G:C]−[A:C]=([G:A]−1)[A:C]≥2,                    (4)
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because A is proper and s_0∉C implies [A:C]≥2.

If u_x∈V_A, an oriented boundary cycle b_x differs from a cycle z supported in X_A by the boundary of a cellular two-chain c over F_p. Across every edge outside X_A which is not a boundary edge of X, the coefficients of the incident outside squares must agree up to the incidence sign, since b_x−z has coefficient zero there. Choose an edge on an outside boundary circle different from xC; its coefficient in b_x−z is zero and it is incident with just one square. The coefficient of that square in c must therefore be zero. Connectivity of the outside face adjacency graph propagates zero to every outside square. But an edge of b_x also has just one incident square, which is outside X_A, and its coefficient in b_x−z is nonzero. This is a contradiction. The argument is valid over every field; it makes no orientability assumption. The proof for B is dual. ■

4. Avoidance of the chosen end components

Choose the lift of 1 corresponding to the identity of H. The chosen lifts of X_A and X_B have vertex sets A_H and B_H. Above a vertex x∈A, the chosen A-component meets the fiber in one affine coset of V_A; above x∉A it has no vertex. Indeed, the differences of two lifted endpoints of A-paths are exactly the homology classes of loops in X_A. The analogous assertion holds for B and V_B. Choosing an origin in each fiber identifies the deck action with translation by V; all these assertions are independent of those origins.

Let E be the induced graph on H(A_H∪B_H). We call a vertex whose image lies in G(A∪B) an exterior-base vertex. The distinction matters: E also contains vertices over A∪B which do not lie in the two chosen components.

Lemma 7. Assume p>4g. Then E is connected.

Proof. A union of two proper subgroups cannot equal a group: if neither contains the other, the product of an element in A\B with one in B\A lies in neither. Thus there is a vertex h∈G(A∪B).

At any exterior-base vertex x, the whole middle circle xC avoids A∪B. All its lifts therefore lie in E. Traversing it changes the deck coordinate by u_x, which belongs to neither V_A nor V_B by Lemma 6. Consequently one can translate any point over x by k u_x, for every k∈F_p, using paths in E.

Consider a fixed base path γ of length L from x, and fix an initial lift. Translate its initial point by k u_x; the entire lifted path is translated by that same deck vector. For any visited vertex over A, the condition that this translated vertex lies in the chosen A-component is an affine equation k u_x∈a+V_A. Since u_x∉V_A, it has at most one solution k. A visited vertex over B excludes at most one further value. Thus at most 2(L+1) values of k are forbidden. Whenever p>2(L+1), one can first travel around the safe middle circle to choose a permitted translate and then lift γ wholly in E.

Every exterior-base vertex reaches some point over h: take a base path to h of length at most g−1 and use the preceding argument. We next show that the entire fiber over h is connected in E. Choose a spanning tree of the base Cayley graph rooted at h. Its fundamental graph loops have length at most 2g−1, and their classes generate H_1(X;F_p), since attaching squares only takes a quotient of graph homology. Let γ be any one of these loops, with class v, and let z be any point over h. There is a permitted k because p>4g≥2(length(γ)+1). Travel from z by k u_h around the middle circle, lift γ, and then travel by −k u_h around the middle circle at h. All three paths lie in E, and their net deck displacement is v. Thus every spanning-tree loop class can be realized between points of the fiber by paths in E. Those classes generate V, so the whole fiber is connected.

It remains to reach an exterior-base vertex from an arbitrary vertex z of E. Let x be its image. If x∈A\B, the edge of color 3 takes x to a vertex outside both A and B: s_3∉A, while s_3∈B and x∉B. The lifted target is therefore automatically allowed. The case x∈B\A is dual, using color 0. If x∈A∩B, first use color 0. The lift remains outside the chosen A-component because it stays in its current A-component, which is unchosen. Its new base vertex xs_0 lies in A\B because s_0∉B, so it is outside the chosen B-component too. Apply the preceding color-3 step. This proves that every vertex of E reaches the already

connected exterior-base region. ■

The field bound is deliberately coarse. No enumeration of H, whose order is given in (2), is required by this argument.

5. Orientability and the two local complements

Let Y_H be the closed surface obtained by filling every middle boundary circle in X_p. Equivalently, it is constructed directly from H by adding 02,03,13 and12 polygons. It satisfies all the hypotheses of the surface construction in phase211. Lemmas 5 and 7 show that

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Y_H\((Y_H)012∪(Y_H)123)                                    (5)
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is connected. We must still treat its orientability and the two smaller selected subcomplexes; connectedness alone is insufficient in odd characteristic for a nonorientable ambient surface.

Lemma 8. If Y_H is nonorientable, the complement in (5) is nonorientable.

Proof. For both the raw and capped surface, orientation transport along a colored edge reverses the local orientation convention determined by the common colored link. Hence the orientation character of a closed graph walk is its length modulo two. In particular X is orientable exactly when its Cayley graph is bipartite. Capping boundary circles does not change orientability, since every middle boundary has even length.

If X is orientable, its cover X_p and its capped surface Y_H are orientable, so only the other case needs consideration. Suppose X is nonorientable. For a spanning tree rooted at h as in Lemma 7, at least one fundamental graph loop γ has odd length; otherwise every graph cycle would have even length. Its length is at most $2g-1$. The construction in Lemma 7 realizes its class v between any point z over h and z+v by a path in E. The two added middle walks have even length, so the constructed path has odd length. Repeat the construction p times, making a new permitted choice of k at each stage. The total displacement is pv=0 in V. Since p is odd, this gives a closed odd-length walk in E. Its orientation character is nontrivial, so E lies in a nonorientable component of the complement. That complement is connected by (5), hence is nonorientable. This argument also shows directly that Y_H is nonorientable in this case. ■

Lemma 9. Each of

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Y_H\((Y_H)01∪(Y_H)12),
Y_H\((Y_H)12∪(Y_H)23)                                      (6)
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is connected. If Y_H is nonorientable, both are nonorientable.

Proof. We treat the first. Its selected vertex set S_v=H01∪H12 lies in A_H. By Lemma 4, the induced graph on H\A_H is connected. Every vertex of A_H\S_v has a color-3 neighbor outside A_H, so the induced graph on H\S_v is connected too.

We verify that its components are the components of the surface complement. For a 02 square, H01 meets its vertices in a 0-edge or not at all, whereas H12 meets them in a 2-edge or not at all. Here Lemma 3 ensures s_2∉H01, and extreme independence ensures s_0∉H12. Removing one such row and column leaves connected outside vertices. A 03 square meets H01 in at most a 0-edge and H12 in at most one vertex. For the latter assertion, neither s_0, s_3 nor s_0s_3 belongs to H12: the first two are excluded by (1), and s_0s_3∈H12⊆B_H would imply s_0∈B_H. Again the remaining vertices are nonempty and connected on the boundary. A 13 square meets each subgroup in a 1-edge or not at all. Their selected edges cannot be opposite rows, because all selected vertices lie in A_H and s_3∉A_H.

Finally a 12 polygon is wholly selected if it is the chosen H12 coset. Otherwise its intersection with H01, when nonempty, is a coset of H01∩H12=H1 by Lemma 3, hence a single boundary edge of color 1. Deleting its two vertices leaves a connected boundary path, since the polygon has at least four vertices.

Thus every outside face has outside vertices connected along its boundary. Every outside edge also has an outside endpoint: a color-1 edge with a selected endpoint is selected; an outside color-0 edge has no H01 endpoint and cannot have both endpoints in H12; an outside color-2 edge has no H12 endpoint and cannot have both endpoints in H01, by the end intersection property; and a color-3 edge cannot have both endpoints in A_H. The connected-link argument of Lemma 5 therefore completes the comparison and proves connectivity. The other complement is dual.

The top complement (5) is contained in each complement (6). If Y_H is nonorientable, Lemma 8 supplies an orientation-reversing loop in (5), hence in each of the connected complements (6). They are therefore nonorientable. ■

6. The second cover and the minimal descendants

For completeness, the surface criterion being used is the following. If Y is a connected closed surface and S a connected proper subcomplex, the map H_1(S;F_p)→H_1(Y;F_p) is injective in either of these situations:

- Y is orientable and Y\S is connected;
- p is odd, Y is nonorientable, and every component of Y\S is nonorientable.

Indeed the exact sequence of the pair reduces the assertion to surjectivity of H_2(Y)→H_2(Y,S). Coefficients on outside faces propagate across outside edges. In the orientable case, one coefficient is free on each component, and the ambient fundamental class supplies their common diagonal. In odd characteristic a nonorientable component forces its coefficient to equal its negative and therefore to vanish. These observations give the stated criterion, including precisely the needed coefficient qualification.

Apply it to Y_H and all three selected subcomplexes in (5)–(6). Lemmas 7–9 show that it applies. In each of the three intersections of the paired subcomplexes, every component is a disk or a single edge: the top pair has one filled 12 disk for each common middle coset, while the local pairs have edges of color 1 or 2. Their first homology is zero. The homological intersection argument from Lemma 3 therefore applies to the mod-p homology cover of Y_H and gives all three equalities

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T012∩T123=T12,   T01∩T12=T1,   T12∩T23=T2.
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The recursive string C-group criterion proves that T is a string C-group. The path presentation of Y_H identifies its fundamental group with the kernel of W(q_H)→H, where q_H is the order of the marked middle product in H and W(q_H)=W/⟨⟨(r_1r_2)^{q_H}⟩⟩. The kernel of its mod-p abelianization is characteristic, so the resulting deck cover is the Cayley complex of a finite marked group T mapping onto H, with elementary abelian p-kernel. Its composite T→H→G consequently has a p-group kernel. This proves Theorem 1, subject to independent review of the complete argument.

Now suppose G is the non-C monodromy group of a finite rank-four polytope P. The standard string C-group/polytope correspondence makes T a finite regular polytopal cover of P. Explicitly, write the flags of P as W/J, where J is a base-flag stabilizer, so K=core_W(J). A regular polytope with normal flag kernel N covers P if and only if N⊆J, equivalently N⊆K. Normality makes this condition independent of the chosen base flag. Thus every regular quotient still covering P has its marked group mapping onto G=W/K. Among its regular quotients still covering P, choose one minimal under covering; finiteness of T ensures its existence. Call its marked group Q_p. The kernel of Q_p→G is a quotient of the p-group ker(T→G), hence is a p-group. It is nontrivial, since otherwise G itself would be a string C-group. Thus |Q_p|/|G| is a nontrivial power of p. Different primes give different group orders, and therefore different regular polytopes. There are infinitely many primes above 4g, proving Corollary 2.

The same argument also separates the universal kernels: for distinct primes p and ℓ, the common quotient over G of the two constructed covers has kernel whose order divides both a p-power and an ℓ-power, so that kernel is trivial. Their kernel product in W is therefore exactly K. This last observation is not needed for the distinct-order proof of the corollary.

7. Attribution, verification, and current limits

The surface intersection criterion, the careful treatment of the hidden common deck kernel, and the passage from p-group covers to finite minimal descendants reuse phase211 and the unfinished raw-surface calculations in phase224_finite_rank_four.md. The newly asserted steps are the boundary membership test (Lemma 6), the large-prime path construction (Lemma 7), and its combination with orientability and the local complements (Lemmas 8–9). Standard graph connectivity, covering-space theory, cellular homology, Mayer–Vietoris, and the string C-group correspondence are used as stated.

The exact program phase256_voltage_check.py checks five explicitly supplied marked groups of orders 32, 96, 96, 216 and 120 at primes 131, 389, 389, 877 and 487, respectively. It computes the raw homology quotient, tests (3) at every base vertex, and checks translated paths for three specified deck coordinates and every fundamental graph loop. Four of these groups fail the original one-stage closed-surface criterion. The order-120 example is A5×C2 and is nonorientable; the program also constructs a closed odd-length exterior walk by repeating the permitted loop 487 times. These checks use symbolic deck coordinates and do not enumerate the enormous first covers. The four other inputs are marked-group diagnostics, with no asserted realization as monodromy groups of actual polytopes.

The Lean file HomologyCoverAvoidance.lean proves five elementary statements: a transverse affine line meets one affine coset at most once; fewer affine constraints than field elements leave a permitted scalar; the specialization to two possible constraints per visited vertex; the path-length inequality; and preservation of odd parity under odd repetition. It uses only Lean's standard logical axioms. The replay also rejects removal of transversality, weakening of the strict field bound, and replacement of odd repetition by even repetition. These formal statements verify the algebra and counting step used in Lemmas 7–8. Cellular topology, the identification of the actual end fibers with affine cosets, and the final polytope correspondence remain written proofs. Neither the finite checks nor these five formal interfaces alone proves Theorem 1.

The unrestricted finite-cover question in rank at least five remains open within this work. This manuscript makes no assertion about new classifications of polytopes or about how much earlier literature it replaces.